

# Detecting Response Styles in Slider Scales

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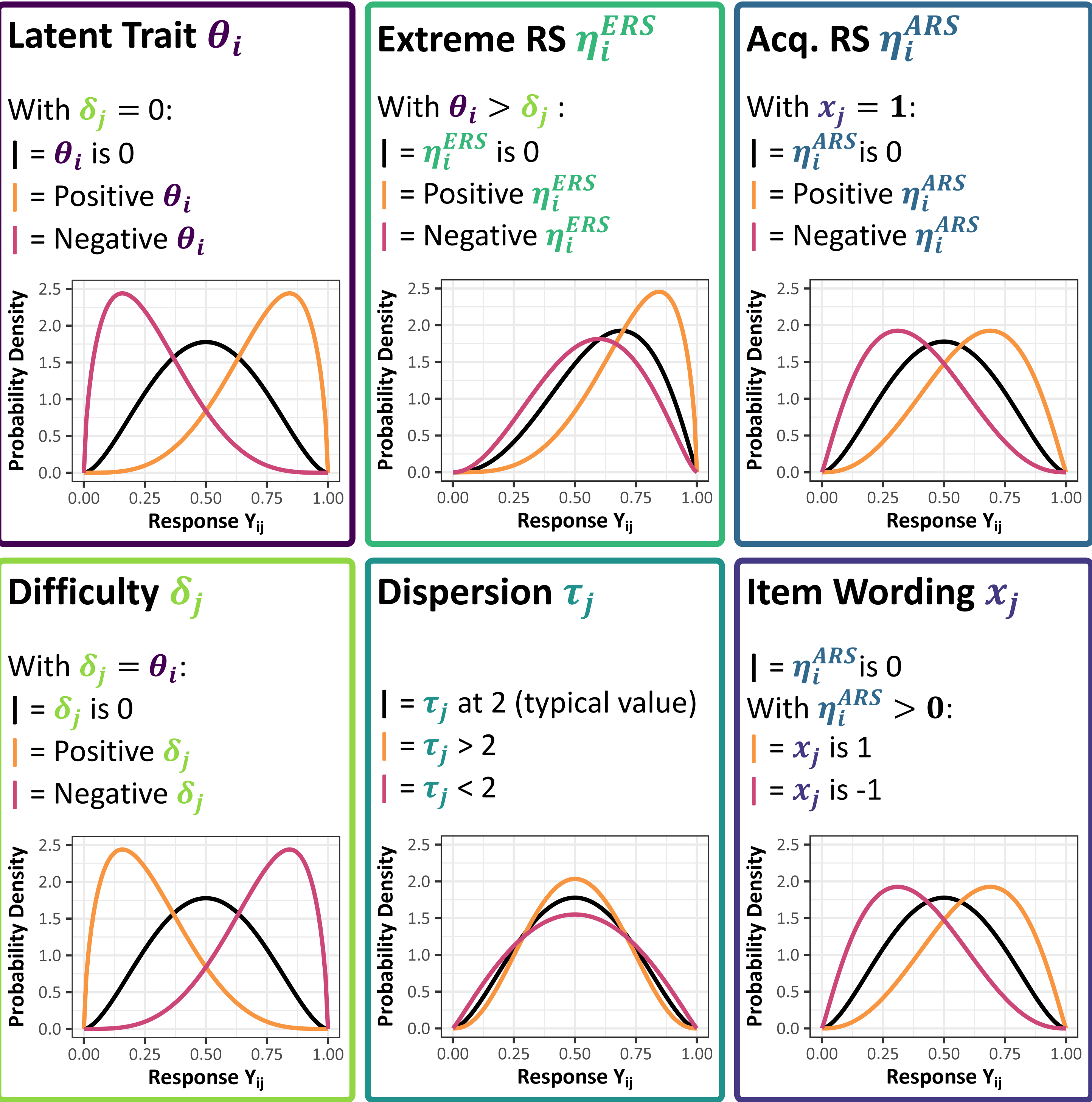
## Introduction: Beta Item Response Model (BIRM)

- Original model by Noel & Dauvier (2007)
- Extended to detect Extreme Response Style (ERS) and Acquiescent Response Style (ARS)

## BIRM With Response Styles (BIRM-RS)

- $i$ : Person,  $j$ : Item

$$m_{ij} = \exp\left(\frac{\exp(\eta_i^{ERS}) \cdot (\theta_i - \delta_j + x_j \cdot \eta_i^{ARS}) + \tau_j}{2}\right),$$
$$n_{ij} = \exp\left(\frac{-\exp(\eta_i^{ERS}) \cdot (\theta_i - \delta_j + x_j \cdot \eta_i^{ARS}) + \tau_j}{2}\right),$$
$$Y_{ij} = \text{Beta}(m_{ij}, n_{ij})$$



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# We developed a model to detect response styles in slider scales.

## Try out the Shiny-App!

## Simulation Study Design

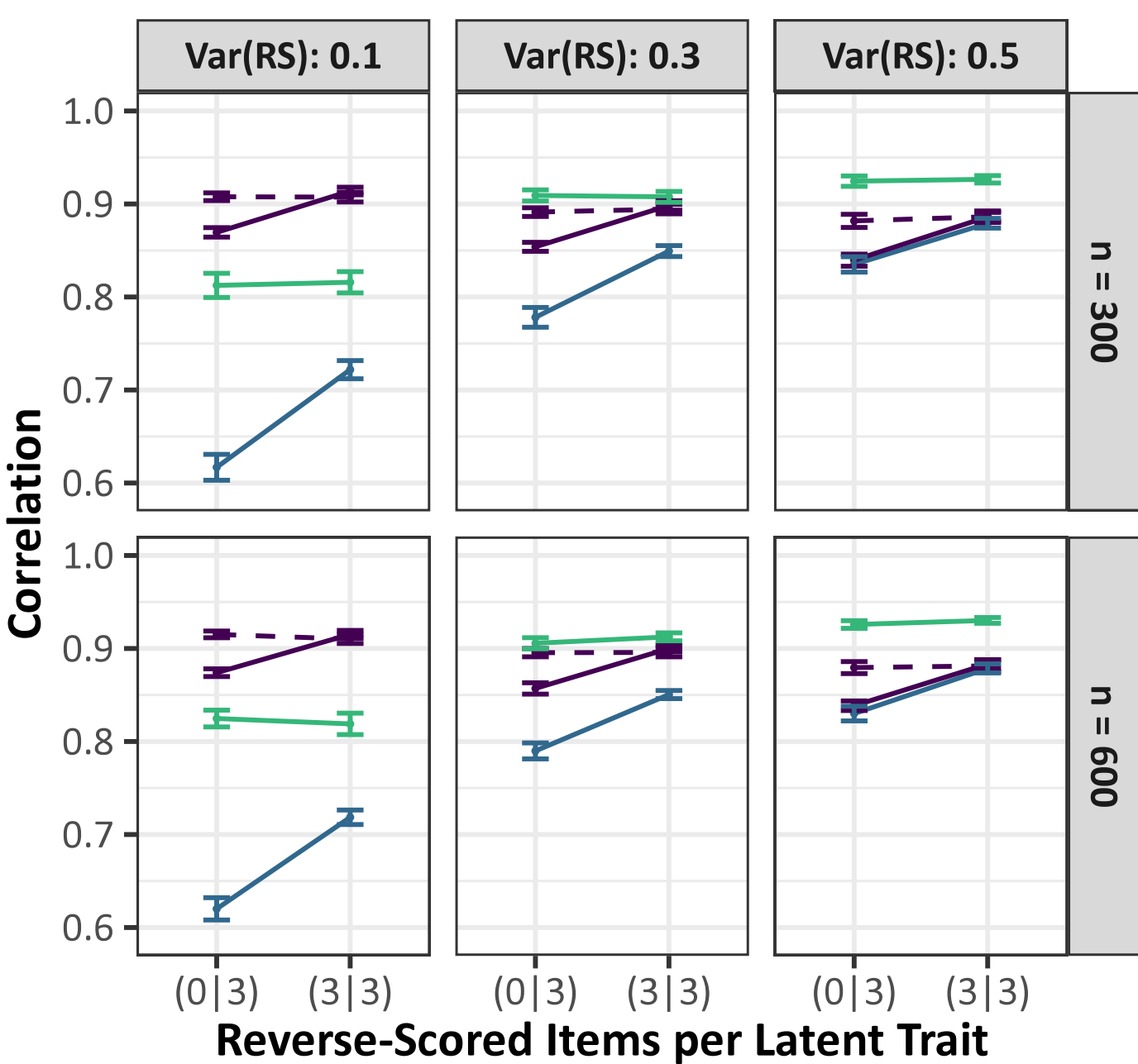
- Goal: Parameter Recovery
- MCMC sampling implemented in Stan
- Two latent traits modeled

## Results

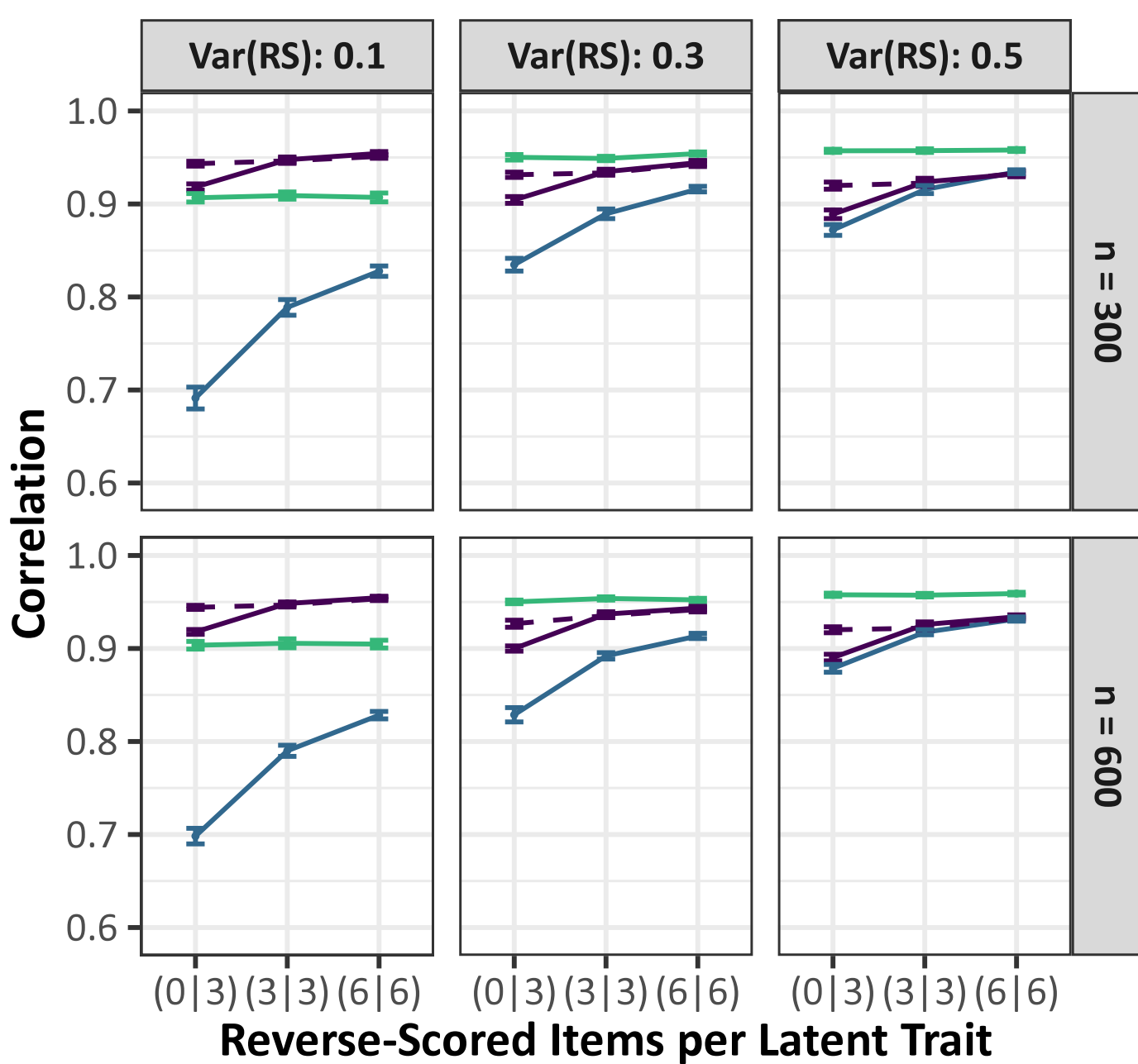
- $\text{—}$  = First latent trait
- $\text{—}$  = Second latent trait
- $\text{—}$  = Extreme RS
- $\text{—}$  = Acquiescent RS

## Correlations Between True and Estimated Values

### 6 Items per Latent Trait

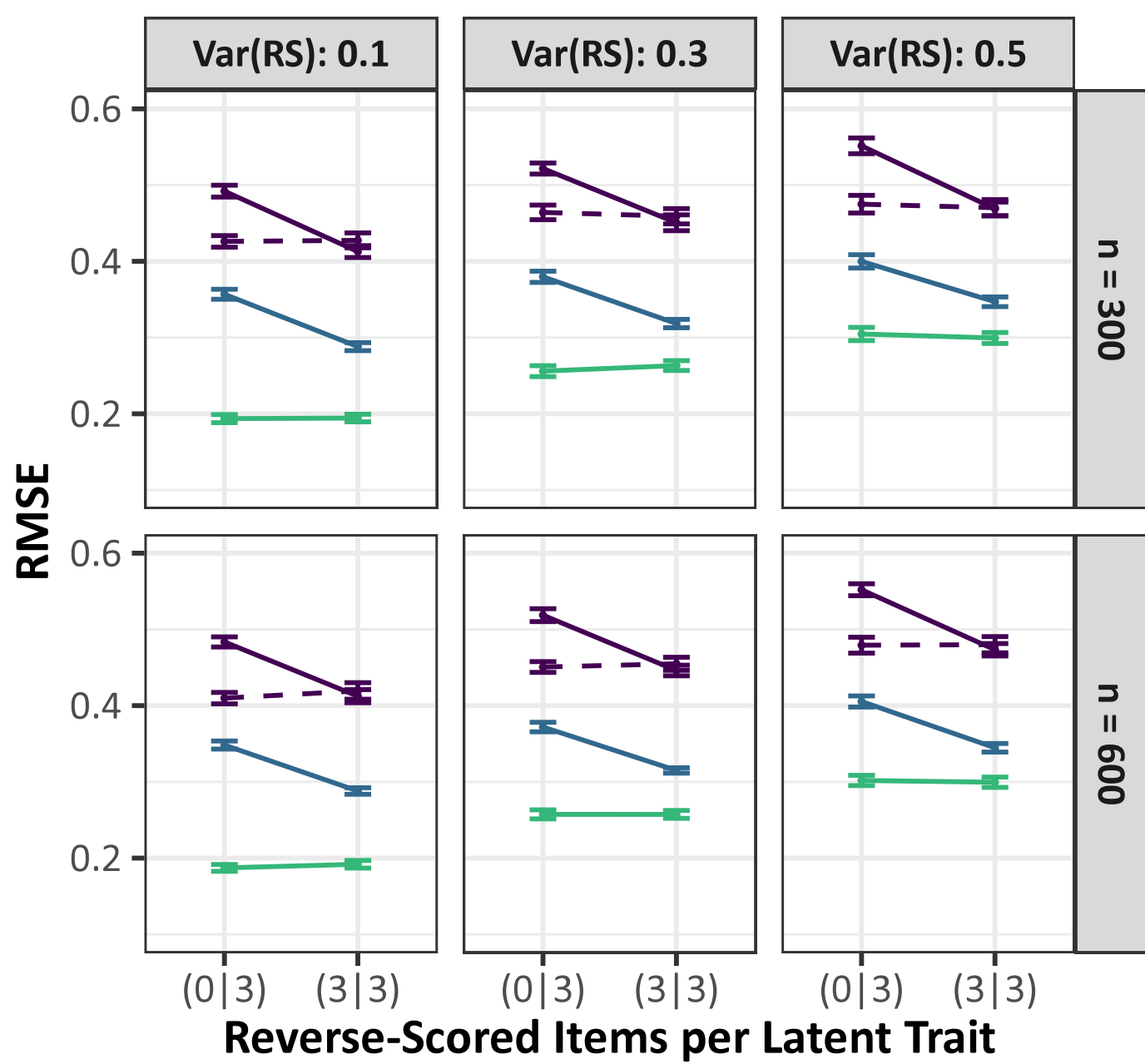


### 12 Items per Latent Trait

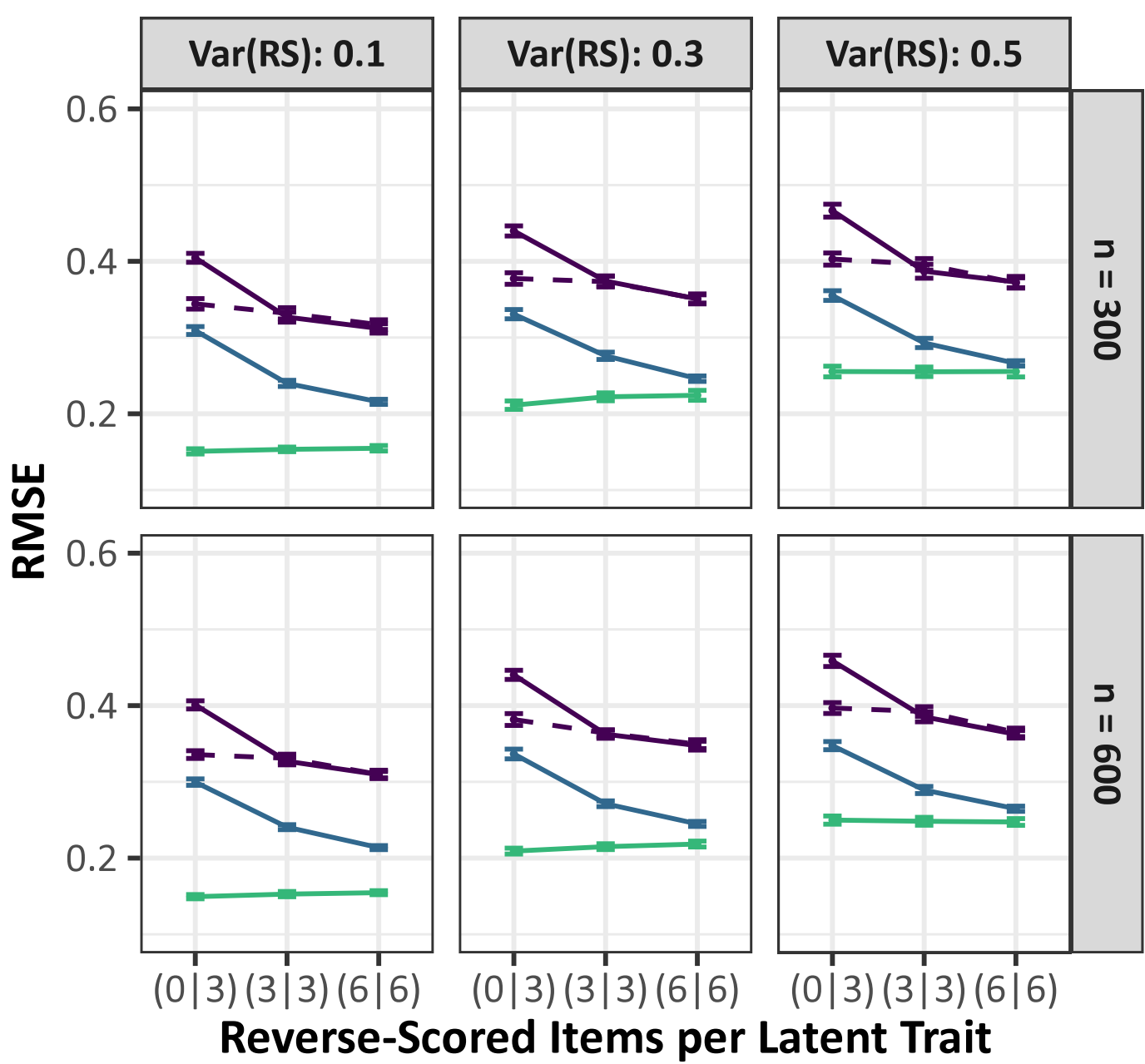


## RMSEs of True and Estimated Values

### 6 Items per Latent Trait



### 12 Items per Latent Trait



## Discussion

- Model successfully recovers parameters given enough (reverse-scored) items and adequate RS variances.
- Empirical illustration planned

## Reference

Noel, Yvonnick, & Dauvier, Bruno (2007). A Beta Item Response Model for Continuous Bounded Responses. *Applied Psychological Measurement*, 31(1), 47–73. <https://doi.org/10.1177/0146621605287691>.